

NAME: _____

PERIOD: _____

DATE: _____

Wrong Shape Does Not Mean Biased

AP Statistics · Unit 3 · Topic 3.2 · B: 2026-11-23 · E: 2026-12-21

[STAR] TODAY'S PROBLEM

Suppose a video platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\hat{p} = 0.14$.

Jules: “Because $n = 100 > 30$, $\hat{p}$ is approximately normal. I get $P(\hat{p} \geq 0.14) = 0.0135$, which proves caption use increased.”

Nora: “Because $np = 8 < 10$, a normal model is not justified. Therefore $\hat{p}$ is biased and this sample tells us nothing.”

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE — before the video (5 min)

Whose reasoning is closer to correct, Jules's or Nora's? Name one condition or claim you would check.

[BOOK] CHECK INDEPENDENCE, THEN SHAPE (keep on the page)

For an SRS, $\mu_{\hat{p}} = p$ and $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$ when $n < 10\%$ of the population makes without-replacement dependence negligible. Approximate normality is a separate question: require both $np \geq 10$ and $n(1-p) \geq 10$; a generic $n > 30$ rule is not enough for proportions. If Large Counts fails, use an exact binomial calculation when independence is reasonable or simulate the sampling process.

Finish after the video:

DOK 1 (a) Identify the population parameter, sample statistic, population size, and sample size. Verify the reported value of $\hat{p}$.

DOK 2 (b) Find the mean and standard deviation of the sampling distribution of $\hat{p}$. Check the 10 percent condition and both Large Counts values.

DOK 3 (c) Adjudicate both claims. Use a binomial calculation or valid simulation for $P(\hat{p} \geq 0.14)$ under $p = 0.08$. State the strongest conclusion warranted, explain why failed normality does not create bias, and give the consequence of each overclaim.

Frame: The 10 percent condition tells us _____, while $np = 8$ tells us _____.

Frame: The result is evidence that _____, but it does not prove _____; Jules misleads by _____.

EXIT — turn this in

One thing the video changed about my first take: _____